

Newton's Method

Newton's method is an iterative numerical technique to find approximations of the roots (or zeros) of a real-valued function. The basic idea is to start with an initial guess x_0 and then use the tangent line at that point to find a better approximation. The formula for Newton's method is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Where:

- x_n is the current approximation,
- x_{n+1} is the next approximation,
- $f(x)$ is the function,
- $f'(x)$ is the derivative of the function.

Problem

a) Find x_3 for $x^3 + 5 = 9x$ using Newton's method starting with $x_0 = 2.75$

1. Initial Values:

$$x_0 = 2.75, \quad f(x_0) = 1.046875, \quad f'(x_0) = 13.6875$$

2. Iteration 1: Compute x_1 :

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2.75 - \frac{1.046875}{13.6875} \approx 2.673515$$

$$f(x_1) = (2.673515)^3 + 5 - 9 \cdot 2.673515 \approx 0.047813$$

$$f'(x_1) = 3 \cdot (2.673515)^2 - 9 \approx 12.44306$$

3. Iteration 2: Compute x_2 :

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.673515 - \frac{0.047813}{12.44306} \approx 2.669673$$

$$f(x_2) = (2.669673)^3 + 5 - 9 \cdot 2.669673 \approx 0.00011837$$

$$f'(x_2) = 3 \cdot (2.669673)^2 - 9 \approx 12.3814$$

4. Iteration 3:

Compute x_3 :

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.669673 - \frac{0.00011837}{12.3814} \approx 2.669663$$

b) Considering the equation $f(x) = x^3 + 4x^2 - 10$, solve the problem using Newton's method considering $x_0 = 1.0$ and find x_4 .

1. Initial Values:

$$x_0 = 1.0$$

$$f(x_0) = 1^3 + 4(1)^2 - 10 = 1 + 4 - 10 = -5$$

$$f'(x_0) = 3(1)^2 + 8(1) = 3 + 8 = 11$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.0 - \frac{-5}{11} = 1.0 + 0.4545455 = 1.4545455$$

2. Iteration 1:

$$x_1 = 1.4545455$$

$$f(x_1) = (1.4545455)^3 + 4(1.4545455)^2 - 10 \approx 1.5401953$$

$$f'(x_1) = 3(1.4545455)^2 + 8(1.4545455) \approx 17.9834711$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.4545455 - \frac{1.5401953}{17.9834711} \approx 1.3689004$$

3. Iteration 2:

$$x_2 = 1.3689004$$

$$f(x_2) = (1.3689004)^3 + 4(1.3689004)^2 - 10 \approx 0.0607197$$

$$f'(x_2) = 3(1.3689004)^2 + 8(1.3689004) \approx 16.5728681$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.3689004 - \frac{0.0607197}{16.5728681} \approx 1.3652366$$

4. Iteration 3:

$$x_3 = 1.3652366$$

$$f(x_3) = (1.3652366)^3 + 4(1.3652366)^2 - 10 \approx 0.0001088$$

$$f'(x_3) = 3(1.3652366)^2 + 8(1.3652366) \approx 16.5135057$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.3652366 - \frac{0.0001088}{16.5135057} \approx 1.3652300$$